

Drought-Proof Generosity: Index Insurance and Informal Livestock Lending in Northern Kenya*

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Abstract

A central concern in the microinsurance literature is whether formal insurance contracts crowd out informal risk-sharing arrangements that underpin livelihoods in developing economies. Using panel data from pastoralist households in Marsabit, northern Kenya, we study how informal networks respond to drought and how Index-Based Livestock Insurance (IBLI) changes that response. We instrument insurance uptake with randomly distributed price discount coupons and interact coverage with sub-location drought severity. Drought erodes informal support for uninsured households. A one-standard-deviation worsening of drought lowers the probability that an uninsured household receives a livestock gift by 11 percentage points and a cash or in-kind transfer by 12 points, and lowers the probability that it gives a transfer by 11 points. Livestock borrowing does not fall. Insured households move in the opposite direction. Our two-stage least squares estimates suggest that the same worsening of drought raises livestock borrowing by roughly 40 percentage points more for insured than uninsured households. In the reduced form, coupon recipients also give and receive more transfers as drought deepens. We find no evidence that insurance crowds out informal risk sharing, and suggestive evidence that it helps keep informal networks active during a drought.

Keywords: index insurance, informal risk sharing, crowding-in, drought, pastoralists, Kenya

JEL Codes: G22, O16, Q12, Q54, D71

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1 Introduction

Informal risk-sharing arrangements such as mutual insurance sustained through gifts, loans, and reciprocal transfers are a foundational feature of consumption smoothing in settings without well-developed financial markets, from village India to rural sub-Saharan Africa (Townsend, 1994; Coate and Ravallion, 1993). Understanding how these networks respond to the arrival of formal insurance in crisis is therefore a question with relevance well beyond any single setting. Households in the arid and semi-arid lands (ASALs) of sub-Saharan Africa offer a particularly sharp context for this. They face severe and recurrent drought shocks with limited access to formal financial markets, and in the absence of insurance rely on a portfolio of informal risk-coping mechanisms; selling livestock at distressed prices, reducing food consumption, and, importantly, borrowing and lending livestock through kinship and social networks (Fafchamps et al., 1998). These informal arrangements provide a crucial, if incomplete, safety net. A fundamental policy question that is relevant to the design of insurance and social protection programs across the developing world is whether the introduction of formal insurance products strengthens or weakens these informal institutions.

On one hand, formal insurance could crowd out informal risk sharing by reducing the need for reciprocal transfers, weakening the social ties that sustain them (Attanasio et al., 2012). On the other hand, insurance could crowd in informal arrangements by making insured households more reliable counterparties—better able to repay and reciprocate—thereby deepening trust and participation in informal networks (Dubois et al., 2008; Mobarak and Rosenzweig, 2012). These two forces need not operate symmetrically on both sides of an informal loan: insurance could change how much a household *borrow*s from its network when it needs help, how much it *lend*s to others in need, or both—with different implications for who ultimately bears risk within the network. Despite the theoretical ambiguity, the empirical evidence is thin, particularly in the pastoral context where the insured object is livestock; the productive asset and the medium of informal exchange.

In this paper, we provide evidence on the causal effect of index-based livestock insurance on informal livestock lending during drought in a pastoral economy. We exploit the IBLI pilot project in Marsabit District, northern Kenya, where pastoralist households have been offered a satellite-based index insurance contract since 2010. The IBLI product compensates policyholders when predicted average livestock mortality, derived from satellite-measured vegetation conditions, exceeds a threshold (Chantararat et al., 2013). A randomly distributed encouragement design provides exogenous variation in insurance uptake that we use to instrument the endogenous decision to insure.

Our empirical strategy differs from the existing IBLI impact literature in two key respects. First, rather than focusing on the widely studied outcomes of asset sales and meal reduction (Janzen and Carter, 2019; Jensen et al., 2017; Bertram-Huemmer and Kraehnert, 2018), we examine a new margin: the household’s participation in informal livestock loan networks. Specifically, we ask whether insured households are more or less likely to borrow livestock from others during periods of drought stress. Second, we exploit the longitudinal structure of the IBLI household survey, whose four coupon rounds (2010 to 2013) identify our main estimates, combined with sub-location-level drought variation derived from the TerraClimate SPEI archive. This allows us to identify the *interaction* of insurance coverage and contemporaneous drought severity within a household fixed effects framework, rather than relying on the single payout event of 2011 that has been the focus of most previous work.

We first document how informal networks behave during drought in the absence of insurance. For uninsured households, a one-standard-deviation worsening of drought lowers the probability of receiving a livestock gift by 11 percentage points, of receiving a cash or in-kind transfer by 12 points, and of giving a

transfer by 11 points. Livestock borrowing and lending do not change. Drought is a covariate shock that hits most members of a network at once, and these results show gift and transfer networks shrinking when households need them most, while the livestock loan network holds steady.

Our main finding is that insured households respond differently. Our estimates suggest that IBLI coverage raises the probability of borrowing livestock as drought worsens, a response absent among uninsured households. The estimated effect is large. A one-standard-deviation worsening in drought severity raises the borrowing probability by roughly 40 percentage points more for insured than uninsured households. This estimate is significant at the 10% level ($p = 0.09$) under both small-cluster CR2 and conventional standard errors, and the reduced form, which relies only on random coupon assignment, is significant at the 5% level. We find no significant effect on livestock lent to others (loan-out) or on livestock received as gifts, and an imprecise estimate for drought-related herd migration that we cannot distinguish from zero. The absence of a lending response among sampled households is not itself evidence against a broader lending response: as we discuss below, the great majority of borrowing partners lie outside our sampled panel, so any additional lending that accommodates this borrowing is, by construction, invisible to a loan-out measure defined only over our roughly 924 sampled households. The null result for gifts, by contrast, is reassuring for our identification strategy: if the discount coupon instrument were spuriously correlated with unobserved social connectedness, we would expect gift receipt to respond as well, and it does not.

Direct evidence on the composition of these lending partners supports this. While our data do not link borrowing events to specific counterparty households, 89% of livestock loans originate from within the household’s own location or sub-location, and relationship data available show these are predominantly family (69%) and clan (22%) members. Because the IBLI panel samples only a small fraction of households within each of Marsabit’s 16 sub-locations, these nearby lending partners are far more likely to be non-sampled pastoralist households than those included in the sample. This explains why the extra borrowing we document among insured households need not appear as extra lending within the panel itself.

We complement the parametric IV estimates with a non-parametric heterogeneity analysis, using the instrumental forest estimator from the generalised random forest (GRF) framework of [Athey et al. \(2019\)](#). This approach lets the data identify which household characteristics matter most for the size of the insurance effect, rather than imposing a single interaction term as in the IV specification above. Drought severity and livestock wealth account for 37% and 34% of splits, respectively, together explaining more of the treatment-effect heterogeneity than all other covariates combined. This wealth-based pattern echoes a broader literature on asset-based poverty traps, in which pastoralist households behave differently depending on whether their herd size lies above or below a critical threshold ([Janzen and Carter, 2019](#); [Carter and Lybbert, 2012](#)). The forest’s formal tests of heterogeneity are not significant, so we treat this pattern as descriptive.

These findings contribute to several strands of the literature. First, we provide evidence on the interaction between formal and informal insurance in a developing economy, bearing on the theoretical ambiguity about crowding out versus crowding in. Our results are consistent with a crowding-in interpretation: by providing a cash or asset buffer during drought, IBLI may enhance households’ perceived creditworthiness within informal networks, or reduce the perceived risk of non-repayment, thereby facilitating livestock loans that would not otherwise occur ([Dubois et al., 2008](#); [Boucher et al., 2009](#)). Suggestive support for this channel comes from the fact that the response appears specific to in-kind livestock loans: the same specification applied to informal cash loans shows no comparable response (coefficient -0.06 , $p = 0.70$). This pattern is more consistent with insurance easing a reputational or repayment-risk constraint specific to reciprocal, relationship-based livestock exchange, where the lender bears the risk that a borrowed animal is not returned

or replaced, than with a general relaxation of household liquidity, which should also raise cash borrowing. Our cash-loan estimates are imprecise and we do not observe lenders’ perceptions directly, so this remains an indirect test of the mechanism rather than a conclusive one; eliciting trust and reputation directly is a natural next step. Second, we contribute to the growing IBLI impact literature by documenting how informal networks contract during drought and by identifying a possible new channel, network-mediated livestock lending, through which insurance could support post-shock resilience. Third, we demonstrate how modern machine learning methods for heterogeneous treatment effects can complement traditional IV approaches in the study of microinsurance impacts.

The rest of the paper proceeds as follows. Section 2 describes the research setting and the IBLI insurance product. Section 3 presents the data and key variables. Section 4 outlines the empirical strategy. Section 5 presents the main results and robustness checks, and Section 5.8 discusses mechanisms. Section 6 concludes.

2 Setting

2.1 Pastoralism in Northern Kenya

Northern Kenya’s arid and semi-arid lands receive low and variable rainfall. Financial infrastructure remains limited, and mobile pastoralism forms the main livelihood system. Marsabit County, formerly Marsabit District, organizes the study area into four sub-counties (North Horr, Saku, Moyale, and Laisamis). These sub-counties divide into 20 electoral wards, which divide further into sub-locations. Our household sample is drawn at the sub-location level (Kirui et al., 2022). The county covers roughly 70,000 km², an area similar in size to the Republic of Ireland, at an overall population density of 6 to 7 people per km² (Kirui et al., 2022). Sub-county size and density vary widely. Saku, which contains Marsabit town, covers 2,078 km² at a density of roughly 22 people per km². Laisamis covers 20,266 km² at a density of roughly 3 people per km². Sub-locations subdivide these sub-counties further, so a single sub-location can cover an area far larger than a typical rural administrative unit elsewhere in Kenya, with a correspondingly low population. Annual rainfall across the county ranges from 200 to 1,000 mm and averages 254 mm, one of the lowest averages among Kenya’s counties (Kirui et al., 2022).

Livestock, including camels, cattle, goats, and sheep, serve as both the main productive asset and the main store of wealth for pastoralist households in this region. Households manage changes in forage and water access across time and location through seasonal herd migration. This strategy requires a minimum herd size to work (Toth, 2015). During severe drought, livestock weaken, productivity falls, and mortality rises. Households that fall below the minimum herd threshold face nonlinear asset dynamics that slow recovery, a pattern consistent with poverty trap dynamics (Lybbert et al., 2004; Barrett et al., 2006; Santos and Barrett, 2011).

Informal livestock transfers form a central part of household risk management in this setting. Households routinely borrow and lend livestock to each other through social and kinship networks. These transactions rebalance herds across species and locations, give access to milk and other livestock products, and help households restock after losses. Unlike fixed-term financial credit, these arrangements typically carry no fixed repayment date. Our survey distinguishes short-term borrowing from long-term borrowing. A short-term loan resembles a milking loan, where the household returns the animal once it stops lactating. A long-term loan can keep an animal, or its offspring, with the borrowing household for years, to be returned or reciprocated only once the lender needs it or the borrower’s own herd recovers. Among the loan events that

record loan type, long-term arrangements outnumber short-term ones by a wide margin (40 of 49 events, 82%), a pattern that matches the reciprocal, open-ended character of kinship-based livestock exchange documented elsewhere in pastoralist East Africa (McPeak, 2004). The survey records loan type in only one round, so we treat this figure as suggestive of typical loan duration rather than a precise estimate.

Each loan event also records how many animals change hands. Borrowing households receive a median of 2 animals per event (mean 2.3, up to 26 in one event). Shoats, meaning goats and sheep, make up 60% of borrowed animals, camels 17%, and cattle 11%. The survey does not link a loan event to the identity of the lending household. We therefore cannot trace a specific loan back to a named counterparty in our panel, and we cannot test reciprocity between two identified households directly. The survey does record where each loan comes from and, in one round, the relationship between the two parties. In Rounds 1 to 5, 89% of the livestock loans with a recorded origin come from within the borrower’s own location or sub-location. In Round 6, family members account for 69% of lenders and clan members for 22%. The lending side looks the same. Of the livestock loans that sampled households made to others with a recorded recipient, 65% went to family members and 18% to clan members (34 events).¹ Because the survey samples between 30 and 100 households per sub-location, most of these nearby kin fall outside our panel. During drought, borrowing takes on particular importance as a coping mechanism distinct from outright purchase or distress sales (McPeak, 2004; Fafchamps et al., 1998).

2.2 The IBLI Insurance Product

Insurers launched the Index-Based Livestock Insurance (IBLI) pilot in Marsabit District in January 2010. Unlike conventional indemnity insurance, IBLI compensates policyholders based on a predicted average livestock mortality index built from satellite-measured normalised difference vegetation index (NDVI) readings, which track vegetative cover as a proxy for rangeland conditions (Chantarat et al., 2013). The index is computed for index areas that group several sub-locations, so every insured household in an index area receives the same payout rate. Our 16 sub-locations fall within four index areas (Central/Gadamoji, Laisamis, Loiyangalani, and Maikona), four sub-locations each. Payouts trigger once predicted mortality exceeds a 15% deductible. Because compensation depends on an exogenous index rather than on individual losses, IBLI avoids the moral hazard and adverse selection problems common in traditional livestock insurance (Jensen and Barrett, 2017; Barnett et al., 2008).

Households choose how many tropical livestock units (TLUs) to insure, and the premium rate varies by geographic region to reflect local risk profiles. Across the 135 Marsabit unit-season records in the program’s sales data (2010 to 2019), the premium rate averages 8.6% of the insured value (interquartile range 7.9 to 9.2%). Uptake fell after the first sales seasons. Total livestock insured in Marsabit fell from 5,965 TLU in 2010 to 413 TLU in 2012, and coverage in our panel fell from 27.6% of households in Round 2 to 9.4% in Round 5. Insured households receive payouts at the end of each dry season, in October or March, if the index triggers. Equity Bank and UAP/APA Insurance sold the product initially, and it has since scaled across Kenya’s drylands and into other countries.

The IBLI research design added a randomised encouragement to boost uptake for evaluation. In each sales period, researchers randomly selected surveyed households to receive discount coupons on the premium for the first 15 TLU insured. In our estimation sample, the share of households receiving a coupon ranges from 57% in Round 2 to 91% in Round 5 (72% overall), and discounts range from 10 to 80% in steps of 10

¹Appendix Table 18 reports the full distribution of livestock loan counterparties in both directions, together with the sources of cash and in-kind transfers.

percentage points. This random assignment gives us exogenous variation in insurance uptake. We observe both the discount percentage a household received and a binary indicator for whether it received any discount at all. Our main specification uses the binary indicator as the instrument, since it does not require us to assume a specific functional form for how discount size maps to uptake.²

3 Data

3.1 Household Panel

Our analysis uses the IBLI Marsabit Household Survey, a panel collected by the International Livestock Research Institute (ILRI) and partner universities. The survey follows a proportionate stratified sample based on the 2009 census. It draws households from 16 sub-locations, with a minimum of 30 and a maximum of 100 households sampled per sub-location. Within each sub-location, the survey draws one third of the sample from each of three tropical-livestock-unit (TLU) wealth strata, low (under 10 TLU), medium (10 to 20 TLU), and high (over 20 TLU), for an initial panel of 924 households. When a household drops out, the survey replaces it with another household from the same wealth stratum and sub-location. This replacement keeps the panel at 924 households through Round 4, 923 in Round 5, and 919 in Round 6 (Kirui et al., 2022).

We use Rounds 1 through 6, spanning 2009 to 2015, for a full panel of 5,530 household-round observations from 1,064 households across the 16 sub-locations. We exclude Round 7 (2020) because the survey did not field the question we use to identify an active IBLI contract that round. We do not read this gap as evidence that national IBLI uptake fell to zero by 2020. Outside our panel, IBLI had scaled commercially across Kenya’s drylands by then. The gap instead reflects a change in the Round 7 questionnaire and the five-year interval since Round 6.

Round 1 predates the first IBLI sales and serves only as a pre-treatment check on our instrument. The discount coupon program that identifies our main instrument ran during Rounds 2 through 5, so our primary estimation sample restricts to these four rounds, giving 3,686 household-round observations from 985 households. Our fixed-effects specifications drop households observed in only one of these rounds, which lowers the estimation sample to 3,652 observations in most of our regression tables. We use Rounds 3 through 6 in a separate specification that checks whether our results hold under a one-round lag in treatment timing.

We build our four main outcome variables from the survey modules on livestock intake, offtake, and herd migration. Section 4 defines each of these variables in the context of our regression model, and Table 21 reports full summary statistics for these variables in our estimation sample.

Livestock borrowing and lending come from separate survey modules, one asking each respondent about animals their own household took in and one asking about animals it sent out. Because each respondent reports only their own household’s side of these transactions, the two rates need not sum to the same total even in a fully reciprocal network. The lending rate also runs lower than the borrowing rate in part because most of a borrowing household’s counterparties fall outside our sampled panel, as Section 2 discusses, so we do not observe the matching increase in lending on their side. We return to this point in Section 5.

Livestock loans are not the only informal arrangement in our data. We also observe general cash and in kind transfers, cash loans, and credit for goods, and we use these to check whether insurance crowds in informal support more broadly or only through livestock loans specifically. General transfers split into

²A first stage that uses the continuous discount percentage in place of the binary indicator produces a similarly strong and significant relationship with IBLI purchase ($t = 6.25$, $N = 3,661$, against $t = 9.79$ for the binary version), so our results do not depend on this choice.

transfers received and transfers given, and the module runs in every round. These transfers also flow through kin and neighbours. Relatives outside the household account for 70% of the transfers households receive and 55% of those they give, and neighbours and friends account for nearly all of the rest (Appendix Table 18). In our estimation sample, a household receives a general transfer in 55.8% of household-round observations and gives one in 43.4%. Informal cash loans run only in Rounds 2 through 5, with a mean of 6.3% in our sample. Formal cash loans, from the same module, have a mean of 2.5%. Informal credit for goods runs in Rounds 4 through 6, with a mean of 49.2% in the two rounds (4 and 5) that overlap our sample. We combine these into two broad indicators. Any informal support received takes the maximum of livestock borrowing, informal cash loans, and informal goods credit, wherever each is observed, with a mean of 37.2%. Any informal support given takes the maximum of livestock lending and general transfers given, with a mean of 46.2%.

We build our treatment variable from the IBLI Contracts. In our primary estimation sample, the IBLI coverage rate is 17.6%. The rate is lower, 12.0%, in the full six-round panel, since it includes Round 1, when no household yet held a policy.

We build our instrument from the discount coupon records. Enumerators distributed these coupons at random, independent of household characteristics. In our estimation sample, 72.2% of households receive some discount. We also observe the size of the discount. Averaged over all households, including those who receive none, the discount averages 29.1% of the premium (SD 24.3). Among the households that receive a discount, it averages 40.3% (SD 19.1). We use this continuous measure as a robustness check.

3.2 Drought Measure

We measure drought severity with the Standardised Precipitation-Evapotranspiration Index at 3-month accumulation (SPEI-3), built from the TerraClimate archive and matched to each sub-location. For each survey round, we take the minimum SPEI-3 value observed during that round’s recall period, which captures the worst drought stress the household faced. In our primary estimation sample, this minimum recall-period SPEI averages -1.073 (SD = 0.560). We center this variable at its sample mean for estimation. Because SPEI varies across sub-locations within an index area, our drought measure is not the same variable that triggers IBLI payouts.

Table 21 in Appendix B reports full summary statistics, number of observations, mean, standard deviation, minimum, 25th percentile, median, 75th percentile, and maximum, for every outcome, treatment and instrument, drought, control, and additional informal-arrangement variable described above, split by TLU wealth stratum (Panel A: low, under 10 TLU; Panel B: medium, 10 to 20 TLU; Panel C: high, over 20 TLU), the same bins used in the original sampling design; given its length we report it in the appendix rather than inline. Livestock borrowing and lending rise with wealth across all three panels, and gift receipt rises from the low to the medium panel, consistent with wealthier households holding a larger and more active position in these exchange networks. IBLI coverage and discount receipt stay close to constant across wealth panels, consistent with the random assignment of coupons.

4 Empirical Strategy

The decision to purchase IBLI is endogenous. Households that choose to insure may differ along unobserved dimensions, including risk preferences, social connectedness, and financial literacy, that also affect their

engagement with informal lending networks. A naive OLS regression of livestock borrowing on insurance status would therefore conflate the causal effect of insurance with selection bias.

We address this endogeneity using the randomly distributed discount coupons as instruments for IBLI uptake, following the encouragement design approach of [Karlan et al. \(2014\)](#) and [Janzen and Carter \(2019\)](#). The key identifying assumption is that coupon receipt affects livestock borrowing only through its effect on the decision to purchase insurance.

Our second stage is

$$y_{it}^S = \beta_1 \widehat{\text{IBLI}}_{it} + \beta_2 (\text{SPEI}_{jt} \times \widehat{\text{IBLI}}_{it}) + \beta_3 \text{SPEI}_{jt} + \mathbf{X}_{it}' \boldsymbol{\phi} + \alpha_i + \gamma_t + \varepsilon_{it} \quad (1)$$

The second stage has two endogenous regressors, IBLI_{it} and $\text{IBLI}_{it} \times \text{SPEI}_{jt}$, so we need two first-stage equations, one for each. We instrument both with coupon receipt, Z_{it} , and its interaction with drought, $Z_{it} \times \text{SPEI}_{jt}$. Each first stage also includes SPEI_{jt} directly, alongside the same controls and fixed effects as the second stage, since SPEI_{jt} also enters the second stage as its own regressor and must therefore appear on the right-hand side of every first stage. Including it lets drought predict insurance uptake on its own, not only through its interaction with the coupon, which matters for households that never receive a coupon. The first stage for the level of coverage is

$$\text{IBLI}_{it} = \delta_0 \text{SPEI}_{jt} + \delta_1 Z_{it} + \delta_2 (\text{SPEI}_{jt} \times Z_{it}) + \mathbf{X}_{it}' \boldsymbol{\theta} + \alpha_i + \gamma_t + v_{it} \quad (2)$$

and the first stage for the interaction term takes the same form, with $\text{IBLI}_{it} \times \text{SPEI}_{jt}$ as the dependent variable and its own coefficients on SPEI_{jt} , Z_{it} , and $Z_{it} \times \text{SPEI}_{jt}$.

IBLI_{it} is a binary indicator for whether household i held an active IBLI contract in round t , our treatment variable. Z_{it} is a binary indicator for whether household i received a discount coupon for the insurance sold in round t , our instrument. SPEI_{jt} is the centred minimum recall-period SPEI-3 value for sub-location j in round t , our measure of drought severity, where a lower value marks worse drought. $\widehat{\text{IBLI}}_{it}$ and $\widehat{\text{IBLI}}_{it} \times \text{SPEI}_{jt}$ are the predicted values of IBLI_{it} and its interaction with drought, taken from the first stage. y_{it}^S denotes the outcome, and we estimate the model separately for each of four binary outcomes built from the survey modules described in Section 3, whether the household borrowed livestock, lent livestock, received a livestock gift, or moved its herd to cope with drought. We treat livestock gifts as a placebo outcome, since gift transfers follow different social norms than reciprocal loans and should not respond systematically to insurance coverage. $\mathbf{X}_{it}$ collects our three time-varying controls, log livestock wealth (using standard TLU conversion factors, cattle 0.7, camels 1.0, goats and sheep 0.1), household size, and settlement status (coded as nomadic, partially settled, or fully settled). α_i and γ_t are household and round fixed effects, which absorb any time-invariant household characteristic and any shock common to all households in a given round. β_2 , the coefficient on the IBLI by SPEI interaction in Equation 1, is our parameter of interest. It captures how insurance changes the household's use of the loan network as drought worsens. Because SPEI is centred and IBLI is binary, β_3 measures how the outcome changes with drought for uninsured households. We use it to describe how informal networks respond to drought in the absence of insurance.

Standard errors are clustered at the sub-location level ($G = 16$). With so few clusters, conventional cluster-robust (CV1) standard errors can over-reject, particularly when cluster sizes are unbalanced ([Mackinnon et al., 2023](#)). In our estimation sample, cluster sizes range from 179 to 621 observations (max to median ratio of 2.23). We therefore report CR2 bias-corrected standard errors ([Pustejovsky and Tipton,](#)

2018) with Satterthwaite-adjusted degrees of freedom as our primary inference method.³

The first-stage relationship between coupon receipt and IBLI uptake is strong. The F -statistic on the binary coupon indicator is 95.8 and the continuous coupon value also predicts uptake ($F = 39.1$). When we instrument the level and the SPEI interaction jointly, the first-stage F -statistics stay above 30.

We regress Round 1 livestock borrowing, which predates the launch of IBLI, on coupon assignments made in later rounds. The estimated coefficient is 0.0003 ($p = 0.62$), which gives no evidence that future coupon receipt predicts pre-treatment outcomes.

The Wu-Hausman test rejects exogeneity of IBLI for livestock borrowing ($p = 0.022$), confirming that the gap between the OLS and 2SLS estimates discussed in Section 5.1 is too large to attribute to sampling noise.

We also check whether the treatment effect varies with herd size. Table 21 shows that livestock borrowing and lending rise with wealth in the raw data, while IBLI coverage and the discount coupon reach each wealth group at a similar rate, consistent with the random assignment of coupons. Section 5 explores this pattern nonparametrically using a generalised random forest, which assigns livestock wealth the second highest weight, after drought severity, in explaining how the treatment effect varies across households.

5 Results

5.1 OLS

Before turning to the instrument, we report what a naive regression of coping strategies on IBLI coverage would show. This regression replaces $\widehat{\text{IBLI}}_{it}$ and $\widehat{\text{IBLI}}_{it} \times \text{SPEI}_{jt}$ in Equation 1 with the household’s actual coverage and its actual interaction with drought, estimated by OLS rather than 2SLS. Because this specification does not need the instrument, it uses a larger sample than the 2SLS models that follow, 5,445 household-round observations across Rounds 1 to 6, rather than 3,652 across Rounds 2 to 5.

Table 1 reports the results. Every coefficient is small and far from significant. The interaction between IBLI and SPEI ranges from -0.009 to 0.015 across the four outcomes, and equals -0.006 for livestock borrowing, two orders of magnitude below the 2SLS estimate in Table 3. The level term on IBLI itself never exceeds 0.021 in absolute value.

IBLI coverage is a choice, not an assignment, and the households who choose to insure without any nudge from a coupon are unlikely to be a random slice of the population. Two features of this setting point to why that choice correlates with the outcomes we study.

First, the same unobserved traits that lead a household to buy insurance on its own probably also shape its position in the informal loan network. A household with strong kinship ties and an active lending relationship already has a form of insurance. It may feel less need to buy IBLI, or, if it does buy IBLI, may not need to lean on the loan network any harder during drought because it never leaned on it lightly to begin with. A household with weak ties may buy IBLI precisely because it lacks that cushion, and may still lack it during drought regardless of coverage. Either pattern pushes the OLS coefficient toward zero, since it averages over households who insure for very different reasons.

Second, OLS and 2SLS do not estimate the same quantity even without this kind of selection. OLS reflects an average across everyone who ends up insured, including households that would have bought IBLI with or without a coupon. The 2SLS estimates in Table 3 isolate the effect for households whose purchase

³All main conclusions hold under either variance estimator.

Table 1: OLS Estimates of IBLI on Coping Strategies

	Loan In	Loan Out	Gift In (Placebo)	Drought Herd
IBLI	−0.004 (0.022)	0.003 (0.014)	−0.020 (0.023)	−0.021 (0.028)
IBLI × SPEI	−0.006 (0.035)	0.001 (0.021)	−0.009 (0.066)	0.015 (0.049)
Household and round FE	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes
Observations	5,445	5,445	5,445	5,445

Notes. OLS regression of each outcome on IBLI coverage, its interaction with centred SPEI, log TLU, household size, and settlement status, with household and round fixed effects. Standard errors clustered by sub-location ($G = 16$) in parentheses. The sample is larger than in Table 3 because this specification does not require the discount instrument to be observed. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

decision the coupon actually changed. If insurance matters most for households who would not otherwise have bought it, the households a coupon is most likely to move, then the two estimates can differ sharply even if neither is wrong on its own terms.

5.2 First Stage

Table 2 reports the first stage. A household that receives any discount coupon becomes 14.6 percentage points more likely to hold IBLI coverage (SE 0.015, $p < 0.001$). This effect equals more than four fifths of the 17.6% coverage rate in our estimation sample. The discount value also predicts uptake. Each additional percentage point of discount raises the probability of coverage by 0.24 percentage points, so the average discount among recipients (40.3%) implies a 9.5 percentage point increase. The F -statistics of 95.8 and 39.1 exceed conventional weak-instrument thresholds by a wide margin. In the full 2SLS model, which instruments both IBLI and its interaction with drought, the first-stage F -statistics equal 46.8 for IBLI and 32.3 for the interaction.

Table 2: First Stage, Coupon Receipt and IBLI Coverage

	(1) Binary coupon	(2) Discount value
Any discount coupon	0.146*** (0.015)	
Discount value (%)		0.0024*** (0.0004)
Household FE	Yes	Yes
Round FE	Yes	Yes
First-stage F	95.8	39.1
Observations	3,661	3,661

Notes. The dependent variable equals one if the household holds an active IBLI contract. Standard errors clustered by sub-location ($G = 16$) appear in parentheses. The sample covers Rounds 2 to 5, when the survey records coupon assignment. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

The coupon does not predict borrowing before IBLI existed. When we regress Round 1 livestock borrowing on the largest discount a household later receives, the coefficient equals 0.0003 per percentage point of discount ($p = 0.62$, $N = 889$). Coupon assignment therefore shows no link to borrowing behaviour that predates the product.

5.3 Insurance and Livestock Borrowing during Drought

Table 3 reports our main estimates for the four outcomes. Panel A shows the reduced form, Panel B shows 2SLS estimates with CR2 standard errors, our preferred inference given only 16 clusters, and Panel C shows 2SLS estimates with conventional CV1 standard errors for comparison. Because a lower SPEI value marks worse drought, a negative coefficient on the interaction means that insured households use a coping strategy more as drought worsens.

Table 3: Effect of IBLI on Coping Strategies as Drought Worsens

	Loan In	Loan Out	Gift In (Placebo)	Drought Herd
<i>Panel A. Reduced form</i>				
SPEI $\times$ any coupon	-0.073** (0.028)	-0.002 (0.023)	-0.025 (0.053)	-0.042 (0.029)
<i>Panel B. 2SLS, CR2 standard errors</i>				
IBLI $\times$ SPEI	-0.710* (0.381)	0.074 (0.202)	-0.584 (0.515)	-0.469 (0.345)
<i>Panel C. 2SLS, CV1 standard errors</i>				
IBLI $\times$ SPEI	-0.726* (0.400)	0.055 (0.225)	-0.600 (0.560)	-0.441 (0.358)
IBLI	0.194 (0.173)	-0.060 (0.099)	0.380 (0.242)	0.134 (0.133)
SPEI	0.015 (0.132)	-0.021 (0.029)	0.191** (0.078)	-0.035 (0.092)
Wu-Hausman p -value	0.022	0.775	0.152	0.263
First-stage F , IBLI	46.8	46.8	46.8	46.8
First-stage F , interaction	32.3	32.3	32.3	32.3
Outcome mean	0.135	0.061	0.334	0.134
Observations	3,652	3,652	3,652	3,652

Notes. All specifications include household fixed effects, round fixed effects, log TLU, household size, and settlement status. SPEI is centred at its sample mean, and lower values mark worse drought. 2SLS models instrument IBLI and IBLI $\times$ SPEI with coupon receipt and coupon receipt $\times$ SPEI. Panel B uses CR2 standard errors (Pustejovsky and Tipton, 2018) with Satterthwaite degrees of freedom; the Satterthwaite df is 9.6 for all four outcomes. Panels A and C use CV1 standard errors. All standard errors are clustered by sub-location ($G = 16$). Coefficient standard errors in parentheses; the Wu-Hausman row reports a diagnostic p -value rather than a coefficient. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Insurance is associated with more livestock borrowing as drought worsens, though the evidence is suggestive rather than conclusive. The reduced form shows that households assigned a coupon borrow more as SPEI falls (coefficient -0.073 , $p = 0.020$). This estimate relies only on random assignment and requires no assumption about who takes up insurance. The 2SLS estimate scales this effect by the first stage. Under CR2 inference the interaction equals -0.710 ($p = 0.094$, Satterthwaite $df = 9.6$), significant at the 10%

level but not at the conventional 5% level. The CV1 point estimate equals -0.726 ($p = 0.090$). The two estimators agree closely in both magnitude and precision. A one standard deviation fall in SPEI (0.56 in our estimation sample) implies an increase in borrowing of roughly 40 percentage points more for insured households than for uninsured households, under either estimator.

This magnitude exceeds the 13.5% average borrowing rate by a wide margin, and we read both the size and the precision of this estimate with care. The estimate measures a local average treatment effect for compliers, the households that buy insurance because they receive a coupon. It also measures a differential response to a large change in drought conditions, not an average effect across all rounds. At average drought conditions, the IBLI coefficient in Panel C equals 0.194 and is not significant, so insurance does not raise borrowing in ordinary years. The effect, such as it is, appears only when drought becomes severe, and with only 16 clusters we cannot estimate it with the precision a larger sample would allow.

The Wu-Hausman test rejects exogeneity of IBLI for livestock borrowing ($p = 0.022$), confirming that the gap between the OLS and 2SLS estimates discussed in Section 5.1 is too large to attribute to sampling noise.

The other three outcomes show no significant response under either standard error estimator. Insurance leaves livestock lending unchanged ($p = 0.723$), and the point estimate is small relative to its standard error. As Section 2 discusses, most of a borrower’s lending partners live in the same sub-location but fall outside our sample. We therefore would not expect to observe the matching rise in lending even if it occurs. Gift receipt serves as a placebo. If the coupon captured unobserved social connectedness, gift receipt should respond as well. The interaction for gifts carries a large standard error and does not reach significance ($p = 0.284$). Drought-related herd movement also shows no significant response ($p = 0.206$), though the point estimate is not small, at -0.469 . We cannot distinguish this from zero, and we also cannot distinguish it from an effect similar in size to the main borrowing result, so we treat this outcome as inconclusive rather than as a clean null.

5.4 Other Informal Arrangements

We next test whether insurance changes other forms of informal support. Table 4 applies the specification in Table 3 to cash loans, goods credit, and general transfers of cash and in-kind goods.

None of these channels shows a significant 2SLS response. Insurance does not raise cash borrowing. The interaction for informal cash loans equals -0.062 ($p = 0.696$), and the interaction for formal loans carries the opposite sign ($p = 0.384$). If insurance only relaxed household liquidity, cash borrowing should rise along with livestock borrowing. It does not, which is consistent with a channel specific to livestock exchange, though the imprecision here means we cannot rule out a similar-sized effect on cash borrowing either.

The reduced form tells a more interesting story than the 2SLS estimates. As drought worsens, households that receive a coupon give more general transfers ($p = 0.035$) and more informal support overall ($p = 0.042$), and receive more general transfers ($p = 0.074$) and more informal support overall ($p = 0.020$). These estimates only require random coupon assignment and no assumption about compliance. Once we scale them by the first stage, the 2SLS estimates for the same outcomes lose significance (p between 0.19 and 0.22), because dividing a marginal reduced-form estimate by an imprecisely estimated first-stage interaction inflates the standard error. Every 2SLS point estimate in the table other than formal cash loans carries the same negative sign as the livestock borrowing result. We read this as a pattern worth following up with more data, not as confirmed evidence that insured households give and receive more.

Table 4: Effect of IBLI on Other Informal Arrangements as Drought Worsens

	Received					Given	
	General Transfer	Informal Cash Loan	Formal Cash Loan	Informal Goods Credit	Any Informal Support	General Transfer	Any Informal Support
<i>Panel A. Reduced form</i>							
SPEI $\times$ any coupon	-0.122* (0.063)	-0.018 (0.015)	0.029 (0.018)	-0.097 (0.124)	-0.117** (0.045)	-0.084** (0.036)	-0.089** (0.040)
<i>Panel B. 2SLS, CR2 standard errors</i>							
IBLI $\times$ SPEI	-0.683 (0.499)	-0.062 (0.153)	0.119 (0.130)	-0.582 (4.122)	-0.634 (0.453)	-0.565 (0.427)	-0.590 (0.424)
Observations	3,652	3,652	3,652	1,822	3,652	3,652	3,652

Notes. Each column reports a separate regression with the controls and fixed effects in Table 3; coefficient standard errors in parentheses. Negative coefficients mean that insured households use the arrangement more as drought worsens. The informal goods credit module covers only Rounds 4 and 5 of our sample, and its first-stage F -statistics (3.8 and 2.0) fall below weak-instrument thresholds, so we do not interpret that column. The Any Informal Support (Received) measure includes livestock borrowing; the Any Informal Support (Given) measure includes livestock lending. Standard errors clustered by sub-location ($G = 16$). Panel B stars use CV1 p -values; none of the Panel B estimates reach the 10% level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

5.5 How Informal Networks Respond to Drought

The models in Tables 3 and 4 also show how informal networks respond to drought among households without insurance. Because SPEI is centred and IBLI is binary, the SPEI coefficient β_3 in Equation 1 measures the change in each outcome as drought worsens for an uninsured household. Table 5 reports this coefficient next to the IBLI by SPEI interaction. A positive SPEI coefficient means the arrangement becomes less common as drought worsens.

Drought shrinks the gift and transfer networks that uninsured households rely on. A one standard deviation worsening of drought lowers the probability that an uninsured household receives a livestock gift by 10.6 percentage points ($p = 0.020$), about a third of the 33.4% mean. It lowers the probability of receiving a general transfer by 12.4 percentage points ($p = 0.025$) and of giving one by 11.1 percentage points ($p = 0.067$). The broad measure of support given falls by 11.3 percentage points ($p = 0.040$). These are the patterns we expect when a covariate shock hits most members of a network at once. Relatives and neighbours who would normally give face the same drought, so there is less to share.

The livestock loan network behaves differently. For uninsured households, drought has no measurable effect on either livestock borrowing ($p = 0.884$) or livestock lending ($p = 0.484$), and both point estimates sit close to zero. Livestock loans are long-term and reciprocal, and the animals remain part of the lender's wealth while they sit in the borrower's herd, so lenders may keep these arrangements in place when they cut back on outright gifts and transfers.

Insured households move against this contraction. For every outcome where drought reduces activity among uninsured households, the IBLI by SPEI interaction is negative and larger in absolute value than the SPEI coefficient. Taken at face value, the point estimates imply that insured households do not experience the contraction in gifts and transfers, and that they increase their livestock borrowing. Only the livestock borrowing interaction reaches the 10% level, so we treat the broader pattern as suggestive. It is consistent with insurance bringing outside resources into a network during a covariate shock, precisely when resources

from inside the network run short.

Table 5: Drought and Informal Network Activity, Uninsured and Insured Households

	SPEI (uninsured)	IBLI $\times$ SPEI
<i>Livestock</i>		
Livestock borrowed	0.019 (0.127)	-0.710* (0.381)
Livestock lent	-0.019 (0.026)	0.074 (0.202)
Livestock gifts received	0.190** (0.067)	-0.584 (0.515)
<i>Cash and in-kind</i>		
General transfer received	0.221** (0.082)	-0.683 (0.499)
General transfer given	0.199* (0.096)	-0.565 (0.427)
<i>Broad measures</i>		
Any informal support received	0.128 (0.101)	-0.634 (0.453)
Any informal support given	0.201** (0.084)	-0.590 (0.424)
Observations	3,652	3,652

Notes. 2SLS estimates of Equation 1. Each row reports the SPEI coefficient and the IBLI $\times$ SPEI interaction from the same regression. CR2 standard errors in parentheses, clustered by sub-location ($G = 16$), with stars based on Satterthwaite degrees of freedom ($df \approx 9.1$ for SPEI and 9.6 for the interaction). The SPEI coefficient measures the effect of drought on uninsured households. A positive SPEI coefficient means the outcome falls as drought worsens. A negative interaction means insured households use the arrangement more than uninsured households as drought worsens. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

5.6 Robustness

Alternative drought measures. Table 6 replaces the recall-period minimum SPEI with three other measures. These specifications instrument IBLI with the continuous discount value rather than the binary coupon indicator. The recall-period mean produces an interaction of -0.430 ($p = 0.013$), which matches the sign of our main estimate. The long dry season average (June to August) also produces a negative interaction (-0.680), though with less precision ($p = 0.222$). The binary drought indicator equals one when SPEI falls below -1 , so a positive interaction marks a larger response in drought. The interaction equals 0.288 ($p = 0.067$). In this specification, insured households borrow 26.0 percentage points less than uninsured households in non-drought rounds ($p = 0.014$), and the gap closes during drought. Across all three, insurance shifts borrowing toward drought periods, with the same sign as our main result.⁴

Timing of coverage. Drought conditions could affect both the purchase decision and borrowing in the same round. To address this concern, we replace current coverage with coverage in the previous round, instrumented by the coupon from the previous round. Table 7 reports the results. The lagged first stage

⁴Instrumenting with the continuous discount value while keeping our main drought measure gives an interaction of -0.803 (CV1 standard error 0.515), close to the -0.726 we obtain with the binary coupon indicator (Appendix Table 11).

Table 6: Robustness to Alternative Drought Measures, Livestock Borrowing

	(1) Recall mean	(2) Long dry season	(3) Binary drought
IBLI $\times$ drought measure	-0.430** (0.152)	-0.680 (0.534)	0.288* (0.146)
IBLI	-0.107 (0.117)	-0.077 (0.126)	-0.260** (0.094)
Household and round FE	Yes	Yes	Yes
Controls	Yes	Yes	Yes
Observations	3,652	3,652	3,652

Notes. 2SLS estimates with CV1 standard errors clustered by sub-location ($G = 16$). Columns 1 and 2 use continuous SPEI measures, where lower values mark worse drought, so a negative interaction means more borrowing in drought. Column 3 uses an indicator equal to one when SPEI falls below -1 , so a positive interaction means more borrowing in drought. Each model instruments IBLI and its interaction with the drought measure using the continuous discount value and its interaction with the same measure. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

remains strong ($F = 98.8$). The lagged interaction for borrowing equals 0.108 ($p = 0.596$, CR2), which carries the opposite sign from our main estimate and comes nowhere close to significance. The lagged specification does not reproduce our main result. IBLI contracts cover a single year, so coverage in the previous round often does not protect the household during the current drought. This is consistent with a borrowing response that depends on active coverage at the time of the shock, though given how far both estimates sit from conventional significance, we do not read much into the comparison beyond that the lagged specification offers no support for the main result.

Table 7: Lagged IBLI Coverage, Rounds 3 to 6

	Loan In	Loan Out	Gift In (Placebo)	Drought Herd
IBLI $_{t-1} \times$ SPEI	0.108 (0.195)	0.200 (0.326)	0.160 (0.350)	0.016 (0.176)
p -value (Satterthwaite)	0.596	0.557	0.661	0.930
Satterthwaite df	7.8	7.8	7.8	7.8
Lagged first-stage F	98.8	98.8	98.8	98.8
Observations	3,522	3,522	3,522	3,522

Notes. The treatment equals IBLI coverage in the previous round, instrumented by coupon receipt in the previous round. CR2 standard errors in parentheses. Controls and fixed effects follow Table 3. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

5.7 Heterogeneity by Baseline Wealth Stratum

Table 21 shows that borrowing rises with herd size, from 10.3% among households with fewer than 10 TLU to 21.1% among households with more than 20 TLU. To test whether the effect of insurance also varies across households, we estimate an instrumental forest (Athey et al., 2019). The forest uses coupon receipt as the instrument and estimates the effect of IBLI on livestock borrowing as a function of drought severity, log livestock wealth, household size, settlement status, and district.

Table 8: Instrumental Forest, Variable Importance

Covariate	Share of splits
SPEI (centred)	0.367
Log TLU	0.338
Household size	0.185
District	0.056
Settlement status	0.054

Notes. Variable importance from an instrumental forest estimated with the `grf` package, clustered by sub-location. The outcome is livestock borrowing, the treatment is IBLI coverage, and the instrument is coupon receipt.

Table 8 reports how often each covariate determines a split in the forest. Drought severity accounts for 37% of splits and livestock wealth for 34%, while household size, district, and settlement status account for the rest. Drought severity leads the ranking, consistent with our parametric finding that the effect of insurance depends on drought, even though that finding is itself imprecisely estimated. When we split households at 15 TLU, households below the threshold show a mean estimated effect of 0.39, against 0.23 for households above it. The best linear projection of the estimated effects onto the covariates yields no significant coefficient ($p > 0.20$ for all), and a forest estimated on 16 clusters has limited power. Given the imprecision of our main result, we treat this subgroup comparison as descriptive. Appendix Table 19 reports the distribution of the estimated effects, the best linear projection, and a rank-weighted test of heterogeneity, none of which detects systematic variation, and Appendix Table 20 reports the estimated effects across the wealth distribution.

The instrumental forest above tells us wealth is the second most important driver of heterogeneity in the estimated effect, but a forest split at an internally chosen threshold cannot on its own tell us whether that difference is large enough to trust. We complement it with two more direct checks, both built on the household’s *baseline* wealth stratum: the original sampling stratum described in Section 3.1, fixed at the household’s first appearance in the panel rather than recomputed every round. We use the baseline stratum rather than contemporaneous TLU because current-round wealth moves with drought and with livestock borrowing itself: 42% of households cross a wealth-tercile boundary at least once across Rounds 2 to 5, so splitting a household fixed-effects panel on a moving target would scatter a household’s own observations across regressions and risks conditioning on an outcome of the very behaviour we study. The baseline stratum avoids both problems; it is fixed at first observation, and every household stays in exactly one group throughout the panel.

Table 9 re-estimates our main specification for livestock borrowing separately within each baseline stratum. All three point estimates carry the same sign as our pooled result, and none individually reaches significance under CR2 or CV1 inference. The instrument is not uniformly strong across strata. The joint first stage for IBLI and its SPEI interaction clears 10 in the Low stratum, but the interaction’s own first stage falls to 8.3 in the Medium stratum and the level term’s first stage falls to 9.9 in the High stratum, so the Medium and High columns should be read with that caveat in mind.

Table 9: Livestock Borrowing by Baseline Wealth Stratum

	Low (<10 TLU)	Medium (10–20 TLU)	High (>20 TLU)
<i>Panel A. Reduced form</i>			
SPEI $\times$ any coupon	−0.069* (0.033)	−0.059 (0.037)	−0.111 (0.083)
<i>Panel B. 2SLS, CR2 standard errors</i>			
IBLI $\times$ SPEI	−0.802 (0.459)	−1.315 (1.537)	−0.636 (0.544)
<i>Panel C. 2SLS, CV1 standard errors</i>			
IBLI $\times$ SPEI	−0.832 (0.520)	−1.361 (1.710)	−0.638 (0.592)
IBLI	0.255 (0.195)	0.526 (0.753)	0.007 (0.332)
SPEI	−0.045 (0.112)	0.053 (0.193)	0.101 (0.191)
Wu-Hausman p -value	0.201	0.386	0.090
First-stage F , IBLI	25.8	13.6	9.9
First-stage F , interaction	12.1	8.3	11.3
Households	348	313	290
Observations	1,329	1,201	1,122

Notes. Outcome is livestock borrowing (loan.in). Baseline wealth stratum is the household’s original sampling stratum, fixed at first observation rather than recomputed each round; 97.5% of households retain a single stratum throughout the panel. Controls, fixed effects, and the CR2 small-cluster correction follow Table 3, applied separately within each stratum. Stars in Panel A use reduced-form p -values; no coefficient in Panel B or Panel C reaches the 10% level in any stratum. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 10 asks the same question with a single, formal test. We interact the IBLI $\times$ SPEI term with Medium and High baseline-wealth dummies in one pooled regression and jointly test whether both interaction terms equal zero, i.e. whether the crowding-in slope differs from the Low-wealth baseline at all. Neither wealth-interaction coefficient is individually distinguishable from zero, and the joint test finds no evidence that the slope differs by wealth ($F = 0.15$, $p = 0.865$). The first stage for the two higher-order interaction terms is weak ($F = 3.3$ and 5.3), so this null is better read as underpowered to detect a difference than as confirmation that none exists, the same caveat the instrumental forest’s own best linear projection already carries.

Table 10: Does the Crowding-In Effect Vary by Wealth? A Pooled Test

	2SLS, CR2 standard errors
IBLI $\times$ SPEI (Low, baseline)	−0.553 (0.987)
IBLI $\times$ SPEI $\times$ Medium	−0.862 (2.154)
IBLI $\times$ SPEI $\times$ High	0.047 (1.514)
Joint test, both wealth interactions = 0 (F)	0.15
p -value (HTZ)	0.865
First-stage F : IBLI $\times$ SPEI	10.6
First-stage F : $\times$ Medium interaction	3.3
First-stage F : $\times$ High interaction	5.3
Household and round FE	Yes
Controls	Yes
Observations	3,652

Notes. Single pooled 2SLS regression on the full estimation sample. IBLI $\times$ SPEI is interacted with Medium and High baseline-wealth dummies (Low is the omitted category); the level terms IBLI, IBLI $\times$ Medium, IBLI $\times$ High, and SPEI $\times$ Medium/High are included but not shown. Medium and High enter only as interactions: as time-invariant household characteristics, their own level effect is absorbed by the household fixed effect. Six endogenous regressors, each instrumented by the corresponding interaction of the discount coupon; CR2 standard errors clustered by sub-location. The joint test uses the HTZ small-sample F approximation (Pustejovsky and Tipton, 2018). First-stage F -statistics are for the full six-instrument system. No coefficient in this table reaches the 10% level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

5.8 Mechanisms: What is driving crowding-in?

Our results point to two patterns. Drought shrinks gift and transfer networks among uninsured households, and insured households appear to borrow more livestock during drought. We discuss four channels and bring several additional pieces of evidence to bear on each, most of which we report in Appendix A. None of these tests settles the question on its own, and several rest on small samples or on a single survey round. Taken together, they favour two of the four channels.

Insurance makes a household a safer borrower. A lender in an informal network extends a loan when it expects the borrower to return or reciprocate it. Coverage protects part of the borrower’s herd value during drought, which could lower the perceived risk that the borrower cannot repay. Formal credit markets work the same way, where insurance relaxes borrowing limits (Boucher et al., 2009). The lenders in our setting are overwhelmingly kin. Family and clan members account for 92% of the livestock loans households receive (Appendix Table 18), which fits evidence that risk-sharing ties form among people who know and trust one another (Attanasio et al., 2012), and who are therefore well placed to observe whether a relative holds a policy.

Three pieces of evidence fit this channel. First, the response is specific to livestock. Insurance does not

raise informal cash borrowing (Table 4), which it should if it worked only by relaxing household liquidity. Livestock loans rest on long-term reciprocal ties, and the lender bears the risk that a borrowed animal is not returned or replaced, so a repayment-risk constraint would bind most tightly there. Second, insurance appears to change how much households borrow, not only whether they borrow. Appendix Table 12 replaces the binary outcome with the total number of animals a household borrows in the round. In the reduced form, a one standard deviation worsening of drought raises the number of animals borrowed by coupon recipients by 0.18 more than by non-recipients ($p = 0.024$), about 41% of the 0.43-animal mean. The corresponding change in the probability of borrowing is about 30% of its mean, so the count rises somewhat more than in proportion, which suggests that insured households are not simply taking on more small loans. The 2SLS estimate has the same sign but is imprecise (-2.11 , $p = 0.224$ under CR2). Third, evidence from a similar setting points the same way. In an IBLI experiment in southern Ethiopia, Takahashi et al. (2019) find weak evidence that a randomly matched peer’s insurance purchase raises a respondent’s willingness to make informal transfers to that peer, which is what we would expect if lenders treat an insured counterparty as a better risk. We do not observe lenders’ perceptions directly, and the number of animals lent by our sampled households does not respond (Appendix Table 12), though, as Section 2 explains, most lenders to our sampled borrowers fall outside the sample.

Informal loans cover the basis risk that insurance leaves behind. IBLI pays out on a predicted area-average mortality rate, not on a household’s own losses. A household whose herd losses exceed what the index predicts remains exposed, and IBLI in Marsabit carries substantial basis risk of this kind (Jensen et al., 2016). Mobarak and Rosenzweig (2012) show that informal risk sharing and index insurance can act as complements for exactly this reason. The network absorbs the idiosyncratic part of a loss, while the index covers the covariate part that the network cannot. Under this channel, insured households turn to kin for animals when their own losses exceed the payout, and kin are more willing to help because the insurer has already absorbed part of the shock. The channel also fits our timing result. Borrowing responds to coverage held during the drought and not to coverage held in the previous round (Table 7), and basis risk only arises for a household that holds a policy when the index is computed. We cannot test this channel directly, because doing so would require matching each household’s livestock losses to the payout it received, and our payout data are too thin to support that comparison.

Insurance brings resources from outside the network. Informal risk sharing works poorly against covariate shocks, because the households that would normally help face the same shock (Fafchamps et al., 1998). Section 5.5 shows exactly this contraction among uninsured households. An IBLI payout arrives from an insurer rather than from a neighbour, so it adds to the resources available within the network at the moment they are scarcest. This channel would explain why every interaction in Table 5 points toward more activity for insured households, though only the livestock borrowing estimate is individually significant.

Two descriptive comparisons speak to this channel. Among ever-insured households in Round 7, those who report having received a payout score better on every food security measure we observe (Appendix Table 15). Their reduced Coping Strategies Index is 12% lower (Maxwell and Caldwell, 2008), and their Household Hunger Scale score is 22% lower (Ballard et al., 2011). They also report fewer days of borrowing food or relying on help from relatives. Payout recipients spend most of the money on food (52% of households) and on buying livestock (38%), and only a handful use it to repay debts or give to others (Appendix Table 16). Among the few households asked whether they changed their drought response because they expected a payment, the most common change is selling fewer animals. Payout recipients also borrow livestock less often in Round 7 (6.8% against 10.9%). This fits the same channel. Once a payout has arrived

and been used to restock, the household needs less from its network. The borrowing response in our main estimates instead reflects coverage held while the drought is underway, before any payout is received. These comparisons are not causal. Payout receipt depends on where a household lives and when it held a policy, and the comparison rests on 44 recipients and 101 non-recipients. In the Round 6 cross-section, households that held coverage in the previous round report a less severe drought shock and a faster recovery, but neither estimate is close to significant ($p = 0.50$ and $p = 0.66$; Appendix Table 14).

Payouts give households resources to reciprocate. Livestock lending depends on expected future reciprocity (Ligon et al., 2002; Coate and Ravallion, 1993), so a payout could let an insured household meet obligations that keep its loans in place. The evidence for this channel is the weakest of the four. The reduced form shows that coupon recipients give more transfers during drought, but the corresponding 2SLS estimate is not significant (Table 4). The spending records point the other way. Only 8% of payout recipients report using the payout to repay debts and 4% to give to others (Appendix Table 16). Among insured households, the few that also belong to a group-based savings or lending scheme borrow livestock far more often than other insured households (46% against 12%; Appendix Table 17), which could reflect a greater capacity to reciprocate, but this comparison rests on 13 household-rounds. We therefore treat this channel as a hypothesis for future work rather than a result we have established.

The survey asks households directly whether they coped with food shortages by relying on assistance from others or by selling livestock, which would speak to all four channels. These questions appear only in Rounds 5 and 6, when we must use lagged coverage and the lagged coupon, and in that subsample the lagged coupon barely predicts coverage (first-stage $t = 1.40$). The resulting estimates are uninformative rather than null (Appendix Table 13). More fundamentally, the survey does not link a borrower to its lender, so we cannot observe how lenders respond to a counterparty’s insurance status.

Taken together, the evidence favours the safer-borrower and outside-resource channels, with basis-risk complementarity consistent with the data but untested. This pattern is broadly consistent with the theoretical prediction of Dubois et al. (2008), who show that formal insurance can strengthen informal risk sharing by easing the commitment problem, and with the absence of crowding out that Takahashi et al. (2019) find for IBLI in southern Ethiopia. Our estimates do not support the concern raised by Arnott and Stiglitz (1991) that formal insurance weakens the incentives that sustain informal arrangements.

6 Conclusion

This paper asks how informal risk-sharing networks behave during drought, and whether formal insurance crowds them out or crowds them in. We study index-based livestock insurance among pastoralist households in Marsabit, Kenya, and use randomly assigned discount coupons to instrument for insurance coverage.

Drought strains these networks. Among uninsured households, a one standard deviation worsening of drought lowers the probability of receiving a livestock gift by 11 percentage points, of receiving a cash or in-kind transfer by 12 points, and of giving a transfer by 11 points. The livestock loan network does not contract. Uninsured households borrow and lend livestock at the same rate in drought as in normal conditions.

We find suggestive evidence that insurance raises livestock borrowing as drought worsens. The same worsening of drought is associated with an increase in the probability of borrowing of roughly 40 percentage points more for insured households than for uninsured households, under two different standard error estimators that agree closely with each other. This estimate reaches significance at the 10% level but not at

the 5% level, and our sample of 16 geographic clusters limits how precisely we can estimate it. The reduced form, which relies only on random assignment, is significant at the 5% level. Insurance shows no significant relationship with livestock lending within our sample or with gift receipt, and the herd movement estimate is imprecise enough that we cannot rule out an effect similar in size to the borrowing result. The main result holds in sign with alternative continuous drought measures and with a binary drought indicator. It does not hold when we lag coverage by one round, consistent with an effect tied to coverage held during the shock, though this comparison itself carries substantial uncertainty.

We do not find significant 2SLS evidence that insurance changes cash borrowing, goods credit, or general transfers. In the reduced form, coupon recipients give and receive more general transfers as drought deepens, and the 2SLS point estimates for these outcomes all point toward insured households avoiding the contraction that uninsured households experience. Most livestock borrowing takes place within the household’s own location and with family and clan members, and most of these partners fall outside our sample, which limits what we can say about lending on their side.

Several limitations apply. First, our main estimate is significant only at the 10% level under both standard error estimators, so it should be read as suggestive of a large effect rather than as confirmation of one. Second, the estimates measure effects for coupon compliers, who may differ from the wider pastoralist population. Third, 16 clusters limit the precision of inference even after the CR2 correction. Fourth, the lagged specification does not reproduce the main result, so any effect appears tied to coverage held during the drought itself, though neither estimate is precise. Fifth, the survey does not identify lending households, so we cannot observe reciprocity between specific pairs, which limits our ability to confirm the reciprocity channel directly. Finally, our evidence on how many animals change hands is limited. The number of animals borrowed rises with coverage as drought worsens in the reduced form, but the 2SLS estimate is imprecise, and a longer panel or more sub-locations would be needed to estimate this margin with confidence.

Taken together, these findings point toward index insurance and informal networks acting as complements rather than substitutes in northern Kenya, though our evidence for this is suggestive rather than definitive. We would not want a reader to take from this paper that formal insurance reliably crowds in informal borrowing. We would want a reader to take that drought erodes informal gift and transfer networks, that the data show no sign of insurance deepening that erosion, and that there is some indication, not yet conclusive, that insurance helps keep networks active when a covariate shock leaves them short of resources. A larger sample, spanning more sub-locations or more rounds, would let future work distinguish these possibilities with more confidence. Future research could also survey both sides of each loan to observe reciprocity directly, match household losses to payouts to test whether informal loans cover the basis risk that insurance leaves behind, and test whether lenders view insured households as more reliable borrowers.

A Supplemental Tables and Figures

This appendix collects results from additional analyses that support, but are not central to, the main results. Section A.1 reports a pre-treatment check on the exclusion restriction. Section A.2 reports our main specification with the continuous discount instrument. Sections A.3 through A.8 report the evidence behind the mechanism discussion in Section 5.8. Sections A.9 and A.10 report additional output from the instrumental forest. Several of these analyses use a single survey round or a small subsample, and we note the limits of each where it applies.

A.1 Pre-Treatment Falsification Test

To assess the validity of the exclusion restriction, we test whether future coupon assignment predicts livestock borrowing in Round 1 (2009), which predates the launch of IBLI. The estimated coefficient on the maximum discount value is 0.0003 with a standard error of 0.0006 ($p = 0.618$), providing no evidence that the instrument predicts pre-treatment outcomes. This supports the assumption that coupon receipt is independent of pre-existing household characteristics that determine loan network participation.

A.2 Continuous Discount Instrument

Table 11 compares our main specification, which instruments coverage with the binary coupon indicator, to the same specification instrumented with the continuous discount value. The two interaction estimates are close in magnitude. The continuous instrument yields a larger standard error, consistent with its weaker first stage (Table 2).

Table 11: Livestock Borrowing, Binary and Continuous Instruments

	(1) Binary coupon	(2) Discount value
IBLI $\times$ SPEI	-0.726* (0.400)	-0.803 (0.515)
IBLI	0.194 (0.173)	0.060 (0.171)
SPEI	0.015 (0.132)	0.042 (0.143)
Household and round FE	Yes	Yes
Controls	Yes	Yes
Observations	3,652	3,652

Notes. 2SLS estimates of Equation 1 for livestock borrowing with CV1 standard errors clustered by sub-location ($G = 16$). Column 1 reproduces Panel C of Table 3. Column 2 instruments IBLI and IBLI $\times$ SPEI with the discount value and its interaction with SPEI. Both specifications use Rounds 2 to 5, the rounds in which the instrument is observed. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

A.3 Intensive Margin: Number of Animals Borrowed and Lent

Table 12 applies the specification in Table 3 to the total number of animals a household borrows or lends in a round, summed across all loan events and set to zero for households with no loan event. These outcomes

are counts, so the coefficients are in animals rather than percentage points. The reduced form for animals borrowed is negative and significant at the 5% level, the same pattern as for the probability of borrowing. The 2SLS estimate has the same sign and is imprecise. The Wu-Hausman test again rejects the exogeneity of coverage ($p = 0.032$). Animals lent show no response.

Table 12: Effect of IBLI on the Number of Animals Borrowed and Lent as Drought Worsens

	Animals borrowed	Animals lent
<i>Panel A. Reduced form</i>		
SPEI $\times$ any coupon	-0.316** (0.126)	0.026 (0.139)
<i>Panel B. 2SLS, CR2 standard errors</i>		
IBLI $\times$ SPEI	-2.113 (1.625)	-0.625 (1.942)
<i>p</i> -value (Satterthwaite)	0.224	0.755
<i>Panel C. 2SLS, CV1 standard errors</i>		
IBLI $\times$ SPEI	-2.152 (1.746)	-0.708 (2.250)
IBLI	-0.026 (1.011)	0.787 (1.502)
SPEI	0.158 (0.369)	0.114 (0.285)
Wu-Hausman <i>p</i> -value	0.032	0.770
First-stage <i>F</i> , IBLI	46.8	46.8
First-stage <i>F</i> , interaction	32.3	32.3
Outcome mean (animals)	0.434	0.276
Observations	3,652	3,652

Notes. Outcomes are the total number of animals borrowed (survey item s6q30) or lent (s6q39) across all loan events in the round. Controls, fixed effects, and instruments follow Table 3. Panel B uses CR2 standard errors with Satterthwaite degrees of freedom ($df = 9.6$). All standard errors are clustered by sub-location ($G = 16$). A negative interaction means insured households borrow or lend more animals as drought worsens. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

A.4 Self-Reported Coping Strategies

In Rounds 5 and 6 the survey asks whether the household coped with food shortage by relying on assistance from others or by selling livestock. In Round 5 these questions were asked only of households that reported a food shortage in the previous 12 months. In Round 6 they were asked of every household. Because the coupon program ended after Round 5, we use coverage and coupon receipt from the previous round, as in Table 7. In this two-round subsample the lagged coupon predicts the lagged interaction term well ($t = 4.16$) but predicts lagged coverage itself only weakly ($t = 1.40$). Table 13 reports the estimates. Both interactions carry large standard errors, and we read them as uninformative rather than as evidence that insurance has no effect on these strategies. The Round 5 filter on self-reported food shortage adds a further concern, since coverage could itself affect whether a household reports a shortage.

Table 13: Lagged IBLI Coverage and Self-Reported Coping Strategies, Rounds 5 and 6

	Relied on assistance from others	Sold livestock
<i>Panel A. Reduced form</i>		
$\text{SPEI} \times \text{coupon}_{t-1}$	-0.026 (0.035)	0.011 (0.051)
<i>Panel B. 2SLS, CR2 standard errors</i>		
$\text{IBLI}_{t-1} \times \text{SPEI}$	0.451 (0.935)	0.456 (1.116)
IBLI_{t-1}	-1.053 (0.970)	-0.266 (1.455)
<i>p</i> -value, interaction (Satterthwaite)	0.646	0.697
Satterthwaite <i>df</i> , interaction	6.2	6.2
First-stage <i>t</i> , coupon_{t-1} on IBLI_{t-1}	1.40	1.40
First-stage <i>t</i> , $\text{coupon}_{t-1} \times \text{SPEI}$ on interaction	4.16	4.16
Outcome mean, Rounds 5–6	0.35	0.59
Observations	1,300	1,300

Notes. Outcomes are survey items s7q8c and s7q8e. Controls and fixed effects follow Table 3. The treatment is IBLI coverage in the previous round, instrumented by coupon receipt in the previous round. First-stage *t*-statistics use CV1 standard errors. Outcome means are computed over all households answering the item in Rounds 5 and 6. Standard errors clustered by sub-location ($G = 16$). A negative interaction means insured households use the strategy more as drought worsens. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

A.5 Self-Reported Shock Severity and Recovery

Round 6 asks households whether a drought affected them in the past year, how severe its effect was on their main activity, and how long they took to recover. Table 14 regresses the two answers on coverage in the previous round among households that report being affected. With a single round, we cannot include household fixed effects, so these are cross-sectional estimates. The 2SLS coefficients on coverage are negative for both outcomes, the sign we would expect if coverage cushions the shock, but neither is close to significant.

A.6 Payout Receipt and Food Security

Round 7 (2020) asks households whether they ever received an IBLI payout, and adds a food security module. Neither can enter our panel estimates. Round 7 has no coupon instrument, and its questionnaire does not identify an active contract. Table 15 instead compares ever-insured households by whether they report having received a payout. The reduced Coping Strategies Index weights five consumption coping behaviours in the past seven days by severity (Maxwell and Caldwell, 2008), and the Household Hunger Scale scores three items on food deprivation in the past 30 days from 0 to 6 (Ballard et al., 2011). Higher values of both mean worse food security. Table 16 reports what payout recipients spent their payouts on, and how households that expected a payment changed their response to drought. These comparisons are descriptive. Payout receipt depends on location and on when a household held a policy, both of which also shape its exposure to drought.

Table 14: Lagged IBLI Coverage, Drought Severity and Recovery, Round 6

	Severity (0–3)	Recovery time (1–5)
<i>Panel A. OLS</i>		
IBLI _{t-1}	0.065 (0.119)	0.069 (0.174)
<i>Panel B. 2SLS, CR2 standard errors</i>		
IBLI _{t-1}	-1.315 (1.852)	-1.250 (2.745)
Log TLU	-0.021 (0.066)	-0.122** (0.053)
<i>p</i> -value, IBLI _{t-1} (Satterthwaite)	0.497	0.660
First-stage <i>F</i>		
Outcome mean	[-] 2.23	[-] 2.42
Observations	638	633

Notes. Sample restricted to Round 6 households that report being negatively affected by drought (survey module S7B). Severity (s7q13) runs from 0 (no effect) to 3 (large effect). Recovery time (s7q14) runs from 1 (within a month) to 5 (more than a year, not fully recovered), so a negative coefficient means faster recovery. Controls are log TLU, household size, and settlement status. The 2SLS model instruments coverage in Round 5 with coupon receipt in Round 5. Panel A reports conventional standard errors; Panel B reports CR2 standard errors clustered by sub-location with Satterthwaite degrees of freedom ($df \approx 8.6$). Outcome means are computed over all drought-affected households. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 15: Food Security and Informal Activity by Payout Receipt, Ever-Insured Households, Round 7

	No payout	Payout received
Reduced Coping Strategies Index	15.9	14.0
Days borrowing food or relying on help	1.54	1.41
Household Hunger Scale (0–6)	1.63	1.27
Livestock borrowed	0.109	0.068
Livestock lent	0.010	0.000
General transfer received	0.455	0.545
General transfer given	0.000	0.000
Households	101	44

Notes. Sample of households that held an IBLI contract in at least one of Rounds 2 to 6 and answered the Round 7 payout question. The reduced Coping Strategies Index uses the standard weights of 1, 2, 1, 3, and 1 on eating less preferred foods, borrowing food or relying on help, limiting portion sizes, restricting adult consumption, and reducing the number of meals. The second row reports the unweighted borrowing item alone. The Household Hunger Scale codes each of its three items as 0 (never), 1 (rarely or sometimes), or 2 (often). Rows below the food security measures are shares of households. No ever-insured household reports giving a general transfer in Round 7.

Table 16: Use of IBLI Payouts and Changes in Drought Response, Round 7

	Households	Share (%)
<i>Panel A. Payout spent on (50 households)</i>		
Food	26	52.0
Livestock purchases	19	38.0
Savings	6	12.0
Veterinary services and vaccination	5	10.0
Health services	4	8.0
Repaying debt	4	8.0
School fees	3	6.0
Fodder	3	6.0
Water for the household	3	6.0
Water for livestock	2	4.0
Gifts or sharing with others	2	4.0
Other	3	6.0
<i>Panel B. Change in drought response because a payment was expected (20 households)</i>		
No change	14	70.0
Sold fewer livestock	5	25.0
Kept children in school	3	15.0
Bought more food on credit	1	5.0
Bought less food on credit	1	5.0

Notes. Panel A uses survey item s15q47l, which records whether IBLI payouts were spent on each listed item. Panel B uses item `ibli_payouts_change_coping`, asked of households that said they changed their drought response because they knew payments were coming. Households can report more than one item, so shares do not sum to 100.

A.7 Group Membership and Informal Activity among Insured Households

Table 17 compares insured household-rounds by whether the household uses a group-based savings or lending scheme. Group members use every form of informal support more often. Only 13 insured household-rounds belong to such a group, too few to support a separate instrumented interaction, so we report the comparison as descriptive.

Table 17: Informal Activity among Insured Households by Savings-Group Membership

	Not a group member	Group member
Livestock borrowed	0.123	0.462
Livestock lent	0.048	0.077
Any informal support received	0.290	0.769
Any informal support given	0.457	0.692
Household-rounds	652	13

Notes. Sample of household-rounds with an active IBLI contract. Group members are households that report a CARE savings and lending group or another informal saving scheme as a saving or lending channel (survey item s14q2). Entries are shares of household-rounds.

A.8 Counterparties in Informal Exchange

Table 18 reports who households exchange with. Panel A covers cash and in-kind transfers, recorded in every round from Round 2. Panel B covers livestock loans. The survey records the relationship to the other party only in Round 6, so Panel B rests on 49 borrowing and 34 lending events.

Table 18: Counterparties in Informal Transfers and Livestock Loans

	Received / Borrowed		Given / Lent	
	Events	Share (%)	Events	Share (%)
<i>Panel A. Cash and in-kind transfers, Rounds 2 to 7</i>				
Relatives outside the household	3,499	70.2	1,909	55.3
Neighbour or friend	1,382	27.7	1,541	44.6
Relatives and neighbours	22	0.4	—	—
Other	78	1.6	3	0.1
<i>Panel B. Livestock loans, Round 6</i>				
Family member	34	69.4	22	64.7
Clan member	11	22.4	6	17.6
Neighbour or friend	3	6.1	4	11.8
Other member of the village	1	2.0	2	5.9

Notes. Panel A uses the transfer source recorded in module S9A. Panel B uses items s6q29a1 (borrowed) and s6q38d (lent), fielded in Round 6 only.

A.9 Instrumental Forest Diagnostics

Table 19 reports additional output from the instrumental forest in Section 5. The distribution of estimated effects has a median of 0.25 and a mean of 0.35, but a few estimates are extreme, ranging from -107.5 to 19.0 . An instrumental forest divides the reduced form by a locally estimated first stage, and where the

coupon barely predicts coverage within a leaf, the resulting estimate can be very large, so we rely on the median and on grouped averages rather than on individual predictions. The best linear projection finds no covariate that predicts the estimated effects, and the rank-weighted average treatment effect is negative and small relative to its standard error, so the forest does not detect systematic heterogeneity.

Table 19: Instrumental Forest, Distribution of Estimated Effects and Tests of Heterogeneity

<i>Panel A. Distribution of estimated effects</i>			
		Value	
Minimum		−107.50	
25th percentile		0.050	
Median		0.247	
Mean		0.354	
75th percentile		0.480	
Maximum		18.99	
<i>Panel B. Best linear projection</i>			
	Coef.	SE	<i>p</i>
Constant	10.223	(9.915)	0.303
Log TLU	−0.802	(1.185)	0.499
Household size	0.650	(0.578)	0.261
Settlement status	−3.494	(2.908)	0.230
SPEI (centred)	6.945	(7.181)	0.334
District	−1.437	(2.093)	0.492
<i>Panel C. Rank-weighted average treatment effect (AUTOC)</i>			
Estimate	−1.548	(2.253)	
<i>Panel D. Mean estimated effect by herd size</i>			
	Below threshold	At or above	
10 TLU threshold	0.348	0.365	
15 TLU threshold	0.395	0.228	

Notes. Output from the instrumental forest described in Table 8, estimated with the **grf** package and clustered by sub-location. The outcome is livestock borrowing, the treatment is IBLI coverage, and the instrument is coupon receipt. Panel B standard errors are cluster- and heteroskedasticity-robust. Panel C ranks households by their estimated effect.

A.10 GRF CATE Profiles by Wealth

Table 20 reports the mean estimated CATE by ventile of livestock wealth, from the instrumental forest. While the estimates are noisy at the bin level, the broad pattern shows positive CATEs across most of the wealth distribution, with somewhat larger effects for households in the middle ranges (roughly 2–8 TLU).

B Data and Descriptive Statistics

This appendix collects material describing the data itself: how each variable used in the main text is constructed from the raw survey and climate modules, full summary statistics on those variables by wealth stratum, and the cluster structure that underlies our preferred standard errors.

Table 20: Mean CATE by Livestock Wealth Ventile

Ventile	Median TLU	Mean CATE	<i>N</i>
1	0.0	0.269	185
2	0.2	0.379	185
3	0.6	0.353	185
4	1.2	0.525	185
5	1.8	0.498	185
6	2.5	0.491	185
7	3.4	0.583	184
8	4.1	0.731	184
9	4.9	0.188	184
10	6.0	0.364	184
11	7.2	−0.046	184
12	8.5	−0.124	184
13	10.0	0.438	184
14	11.7	0.618	184
15	13.7	0.478	184
16	16.2	0.461	184
17	19.0	0.120	184
18	24.2	0.105	184
19	33.2	0.290	184
20	55.6	0.363	184

Notes: Estimated from an instrumental forest using `grf`. The treatment is IBLI, the instrument is coupon receipt, and the outcome is livestock borrowing. Clusters are sub-locations. TLU is the median within each ventile.

B.1 Variable Construction

B.1.1 Outcome Variables

Livestock borrowed (`loan_in`). Constructed from Section 6D (Livestock Intake). A household-round observation is coded as 1 if any intake event in that round is classified as `s6q29 == "Loaned in/Borrowed"`.

Livestock lent (`loan_out`). Constructed from Section 6E (Livestock Offtake). Coded as 1 if any offtake event is classified as `s6q38 == "Loaned"`.

Livestock gifts received (`gift_in`). From Section 6D, coded as 1 if any intake event is `s6q29 == "Received as gifts"`.

Drought herd movement (`drought_herd`). From Section 10 (Herd Migration), coded as 1 if any satellite camp has reason `s10q6 == "for coping with drought"`.

B.1.2 Treatment and Instrument

IBLI coverage (`ibli`). From Section 15B (IBLI Contracts). Coded as 1 if the household has at least one non-missing contract record (`s15q19` non-missing) in the round.

Discount coupon (`any_discount`). From Section 15E (Game and Discount Coupon). Mapped to the survey round based on the sales period. The maximum discount value across relevant sales windows is used. Coded as 1 if any discount value > 0 .

B.1.3 Drought Measure

SPEI-3 values are obtained from TerraClimate at the sub-location level. For each survey round, the recall period is defined as October of the prior year through September of the survey year (matching the survey's recall window). The minimum SPEI-3 value over this period captures peak drought stress. The variable is centred at its sample mean (-1.073) for estimation.

B.1.4 TLU Conversion

Livestock holdings from Section 6A are converted to tropical livestock units using the following weights: camels = 1.0, cattle = 0.7, goats = 0.1, sheep = 0.1. The sum across all species gives household TLU, which enters the estimation in logs ($\log(1 + \text{TLU})$).

B.2 Full Summary Statistics by Wealth Stratum

Table 21: Summary Statistics by Wealth Stratum

	Obs.	Mean	St. Dev.	Min	Pctl(25)	Median	Pctl(75)	Max
Panel A. Low Wealth (<10 TLU)								
<i>Outcomes</i>								
Livestock borrowed (loan in)	2,303	0.102	0.303	0.000	0.000	0.000	0.000	1.000
Livestock lent (loan out)	2,303	0.037	0.190	0.000	0.000	0.000	0.000	1.000
Livestock gifts received	2,303	0.260	0.439	0.000	0.000	0.000	1.000	1.000
Drought herd movement	2,303	0.120	0.325	0.000	0.000	0.000	0.000	1.000
<i>Treatment and Instrument</i>								
IBLI coverage	2,303	0.175	0.380	0.000	0.000	0.000	0.000	1.000
Discount coupon received (any)	2,303	0.717	0.450	0.000	0.000	1.000	1.000	1.000
Discount value, all households	2,303	28.8	24.2	0.0	0.0	30.0	50.0	80.0
Discount value, recipients only	1,652	40.2	18.9	10.0	20.0	40.0	60.0	80.0
<i>Drought</i>								
SPEI-3 recall minimum	2,303	-1.090	0.556	-2.179	-1.420	-1.111	-0.627	0.033
<i>Controls</i>								
Livestock (TLU)	2,303	3.6	2.9	0.0	0.9	3.1	5.9	9.9
Household size	2,303	5.6	2.5	1.0	4.0	5.0	7.0	19.0
<i>Additional Informal Arrangements</i>								
General transfer received	2,303	0.591	0.492	0.000	0.000	1.000	1.000	1.000
General transfer given	2,303	0.417	0.493	0.000	0.000	0.000	1.000	1.000
Informal cash loan received	2,303	0.051	0.221	0.000	0.000	0.000	0.000	1.000
Formal cash loan received	2,303	0.034	0.182	0.000	0.000	0.000	0.000	1.000
Informal goods credit received	1,182	0.459	0.499	0.000	0.000	0.000	1.000	1.000
Any informal support received (broad)	2,303	0.338	0.473	0.000	0.000	0.000	1.000	1.000
Any informal support given (broad)	2,303	0.436	0.496	0.000	0.000	0.000	1.000	1.000
Panel B. Medium Wealth (10 to 20 TLU)								
<i>Outcomes</i>								
Livestock borrowed (loan in)	784	0.171	0.377	0.000	0.000	0.000	0.000	1.000
Livestock lent (loan out)	784	0.092	0.289	0.000	0.000	0.000	0.000	1.000
Livestock gifts received	784	0.464	0.499	0.000	0.000	0.000	1.000	1.000
Drought herd movement	784	0.135	0.342	0.000	0.000	0.000	0.000	1.000
<i>Treatment and Instrument</i>								
IBLI coverage	784	0.181	0.385	0.000	0.000	0.000	0.000	1.000
Discount coupon received (any)	784	0.745	0.436	0.000	0.000	1.000	1.000	1.000
Discount value, all households	784	30.2	24.5	0.0	0.0	30.0	50.0	80.0
Discount value, recipients only	584	40.5	19.6	10.0	20.0	40.0	60.0	80.0
<i>Drought</i>								
SPEI-3 recall minimum	784	-1.013	0.591	-2.179	-1.431	-0.952	-0.544	0.033
<i>Controls</i>								
Livestock (TLU)	784	14.2	2.8	10.0	11.8	13.9	16.6	20.0
Household size	784	6.2	2.4	1.0	4.8	6.0	8.0	15.0
<i>Additional Informal Arrangements</i>								
General transfer received	784	0.506	0.500	0.000	0.000	1.000	1.000	1.000
General transfer given	784	0.444	0.497	0.000	0.000	0.000	1.000	1.000

Continued on next page

Table 21 (continued)

	Obs.	Mean	St. Dev.	Min	Pctl(25)	Median	Pctl(75)	Max
Informal cash loan received	784	0.052	0.223	0.000	0.000	0.000	0.000	1.000
Formal cash loan received	784	0.014	0.118	0.000	0.000	0.000	0.000	1.000
Informal goods credit received	434	0.571	0.495	0.000	0.000	1.000	1.000	1.000
Any informal support received (broad)	784	0.434	0.496	0.000	0.000	0.000	1.000	1.000
Any informal support given (broad)	784	0.482	0.500	0.000	0.000	0.000	1.000	1.000
Panel C. High Wealth (>20 TLU)								
<i>Outcomes</i>								
Livestock borrowed (loan in)	600	0.212	0.409	0.000	0.000	0.000	0.000	1.000
Livestock lent (loan out)	600	0.110	0.313	0.000	0.000	0.000	0.000	1.000
Livestock gifts received	600	0.450	0.498	0.000	0.000	0.000	1.000	1.000
Drought herd movement	600	0.183	0.387	0.000	0.000	0.000	0.000	1.000
<i>Treatment and Instrument</i>								
IBLI coverage	600	0.177	0.382	0.000	0.000	0.000	0.000	1.000
Discount coupon received (any)	600	0.708	0.455	0.000	0.000	1.000	1.000	1.000
Discount value, all households	600	28.4	24.4	0.0	0.0	30.0	50.0	80.0
Discount value, recipients only	425	40.1	19.2	10.0	20.0	40.0	60.0	80.0
<i>Drought</i>								
SPEI-3 recall minimum	600	-1.087	0.527	-2.179	-1.514	-1.079	-0.619	0.033
<i>Controls</i>								
Livestock (TLU)	600	39.4	23.5	20.1	24.5	32.0	46.5	270.6
Household size	600	5.9	2.4	1.0	4.0	6.0	7.0	13.0
<i>Additional Informal Arrangements</i>								
General transfer received	600	0.497	0.500	0.000	0.000	0.000	1.000	1.000
General transfer given	600	0.487	0.500	0.000	0.000	0.000	1.000	1.000
Informal cash loan received	600	0.123	0.329	0.000	0.000	0.000	0.000	1.000
Formal cash loan received	600	0.003	0.058	0.000	0.000	0.000	0.000	1.000
Informal goods credit received	231	0.506	0.501	0.000	0.000	1.000	1.000	1.000
Any informal support received (broad)	600	0.418	0.494	0.000	0.000	0.000	1.000	1.000
Any informal support given (broad)	600	0.537	0.499	0.000	0.000	1.000	1.000	1.000

Notes. Sample includes the 3,687 household-round observations in the primary estimation sample, Rounds 2 to 5 (2010 to 2013), the rounds in which the discount coupon instrument is observed. Wealth strata follow the sampling strata (Section 3.1: low <10 TLU, medium 10 to 20 TLU, high >20 TLU), measured contemporaneously in each household-round; because herd size changes over time, the panels here are not equal in size, unlike the original stratified sample at baseline. All binary/indicator variables are reported as proportions (0 to 1) rather than percentages; Obs. gives the number of non-missing household-round observations for that variable within the panel. Discount value is measured in percentage points of the insurance premium (0 to 80); the recipients-only row conditions on a nonzero discount. Livestock (TLU), household size, and SPEI-3 retain their natural units. The informal cash-loan and formal cash-loan modules are fielded in Rounds 2 to 5 (fully covered here); the informal goods-credit module (Section 14D) is fielded in Rounds 4 to 6, so within this Rounds 2–5 sample it is observed only in Rounds 4 and 5, which is why its Obs. column is smaller than the other rows. TLU conversions: cattle 0.7, camels 1.0, goats and sheep 0.1.

B.3 Cluster Diagnostics

The estimation sample spans 16 sub-locations. Table 22 summarises the distribution of cluster sizes.

Table 22: Cluster Size Distribution

	Min	1st Quartile	Median	Mean	Max
Observations per sub-location	179	231	279	346	621

The max/median ratio of 2.23 indicates moderate heterogeneity in cluster sizes. Following [MacKinnon et al. \(2023\)](#), this level of imbalance combined with only 16 clusters motivates the use of CR2 bias-corrected standard errors with Satterthwaite degrees-of-freedom adjustments. The effective degrees of freedom for the main $\text{IBLI} \times \text{SPEI}$ coefficient is 9.6 under the Satterthwaite correction.

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